# The volume of a sphere and of its parts

Four thirds pi r cubed — where it comes from, and how to measure a cap, a segment and a sector.

LESSON 45–46 | 2 × 40 MIN | GEOMETRY 11, §20 | IGCSE E13.X

LESSON CLOCK

14' | 24' | 24' | 22' | 20' | 6'

| | |
|---|---|
| The formula | 14 min |
| Where it comes from | 24 min |
| Caps and hemispheres | 24 min |
| Displacement | 22 min |
| Practice | 20 min |
| Homework | 6 min |

LEARNING OBJECTIVES

- State and use the volume of a ball.
- Derive the formula by Cavalieri or by integration.
- Compute the volume of a spherical cap and of a hemisphere.
- Solve problems on floating, displacement and compound solids.

TEXTBOOK

| | |
|---|---|
| National | pp. 223–234 |
| Cambridge | Core & Extended, pp. 355–360 |
| Workbook | Exercise 20.1 |

## 01 Explanation

### The formula

$$V = \frac{4}{3}\pi R^3$$

| $R$ | $V$ | APPROXIMATELY |
|---|---|---|
| 1 | $\frac{4\pi}{3}$ | 4.19 |
| 3 | $36\pi$ | 113 |
| 6 | $288\pi$ | 905 |
| 10 | $\frac{4000\pi}{3}$ | 4189 |

CUBE THE RADIUS, NOT THE DIAMETER

> A ball of **diameter** $6$ has $R = 3$ and volume $36\pi$, not $288\pi$. Halving first is the commonest error, and it is a factor of eight.

## Where the formula comes from

**By Cavalieri.** Compare a hemisphere of radius $R$ with a cylinder of radius and height $R$ from which a cone has been removed. At height $y$:

$$\text{hemisphere slice} = \pi(R^2 - y^2)\ ;\ \text{cylinder-minus-cone slice} = \pi R^2 - \pi y^2$$

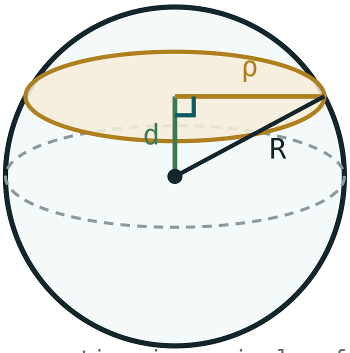
The slices are equal at every height, so the volumes are equal:

$$V\_hemi = \pi R^3 - \frac{1}{3}\pi R^3 = \frac{2}{3}\pi R^3 \Rightarrow V\_ball = \frac{4}{3}\pi R^3$$

**By integration.** Rotating $y = \sqrt{R^2 - x^2}$ about $Ox$ on $[-R, R]$ gives $\pi\int(R^2 - x^2)dx = \frac{4}{3}\pi R^3$ — the same answer, from Quarter III algebra.

## Caps, hemispheres and sectors

| PART | VOLUME |
|---|---|
| hemisphere | $\frac{2}{3}\pi R^3$ |
| cap of height $h$ | $\frac{\pi h^2}{3}(3R - h)$ |
| spherical sector (cap + cone to the centre) | $\frac{2}{3}\pi R^2 h$ |
| layer between two parallel planes | difference of two caps |




*The cap sits above the cutting plane; its height h is measured from the plane to the top.*

**Check the cap formula.** Put $h = R$: $\frac{\pi R^2}{3}(3R - R) = \frac{2}{3}\pi R^3$ — the hemisphere ✓.

Put $h = 2R$: $\frac{4\pi R^2}{3}(R) = \frac{4}{3}\pi R^3$ — the whole ball ✓.

## Displacement

ARCHIMEDES

A solid fully submerged displaces its own volume of liquid. A floating solid displaces its own **weight** of liquid.

**Example.** A ball of radius $3$ cm is dropped into a cylinder of radius $5$ cm containing water. By how much does the level rise?

$$36\pi = \pi(25)\Delta h \Rightarrow \Delta h = \frac{36}{25} = 1.44 \text{ cm}$$

This is how an irregular solid's volume is measured, and it is the practical purpose of the formula.

## 02 Worked examples

**EXAMPLE 1** **Find the volume of a ball of diameter *12* cm.**

1. $R = 6$

   Halve the diameter first.

2. $V = \dfrac{4}{3}\pi(216)$

3. $= 288\pi \approx 905$ cm³.

ANSWER

$288\pi \approx 905$ cm³

**EXAMPLE 2** **A cap of height *4* is cut from a sphere of radius *10*. Find its volume.**

1. $V = \dfrac{\pi(16)}{3}(30 - 4)$

2. $= \dfrac{16\pi}{3} \times 26$

3. $= \dfrac{416\pi}{3} \approx 436$

ANSWER

$\dfrac{416\pi}{3} \approx 436$

EXAMPLE 3 **A ball of radius $3$ is dropped into a cylinder of radius $5$. Find the rise in the water level.**

1. Ball volume $36\pi$.
2. $\pi(25)\Delta h = 36\pi$
3. $\Delta h = 1.44$ cm.

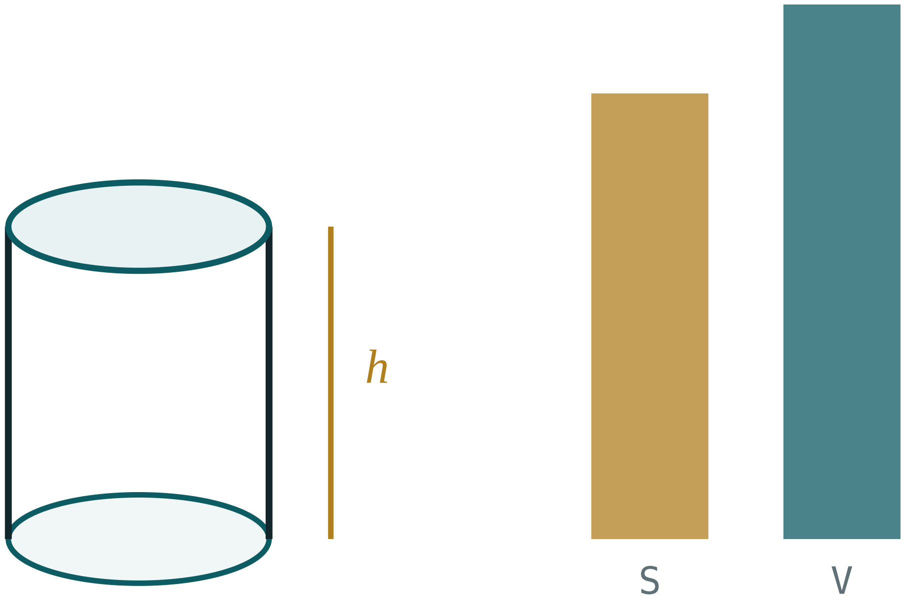
ANSWER

$1.44$ cm

## 03 Interactive model

Drop a ball into a measuring cylinder and read the rise before computing it.

### The sphere against its cylinder




base area B **50.3** | perimeter P **25.1** | total S **251.3** | volume V **301.6**

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH | O‘ZBEKCHA | РУССКИЙ |
|---|---|---|
| Ball | Shar | Шар |
| Hemisphere | Yarim shar | Полушар |
| Spherical cap | Sharsimon segment | Шаровой сегмент |
| Spherical sector | Sharsimon sektor | Шаровой сектор |
| Spherical layer | Sharsimon qatlam | Шаровой слой |
| Displacement | Siqib chiqarish | Вытеснение |
| Cavalieri’s principle | Kavaleri prinsipi | Принцип Кавальери |
| Height of a cap | Segment balandligi | Высота сегмента |
| Archimedes’ theorem | Arximed teoremasi | Теорема Архимеда |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

# 05 Quick check

### Check yourself

The volume of a ball is:

- $4\pi R^2$
- $\frac{4}{3}\pi R^3$
- $\frac{2}{3}\pi R^3$
- $\pi R^3$

A ball of diameter 6 has volume:

- $36\pi$
- $288\pi$
- $72\pi$
- $108\pi$

A hemisphere of radius $R$:

- $\frac{4}{3}\pi R^3$
- $\frac{2}{3}\pi R^3$
- $\frac{1}{3}\pi R^3$
- $2\pi R^3$

A sphere is what fraction of its enclosing cylinder?

- $\frac{1}{2}$
- $\frac{2}{3}$
- $\frac{3}{4}$
- $\frac{1}{3}$

A submerged solid displaces:

- its weight
- its volume
- half its volume
- nothing

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

### Easy · warm the idea up — 7 PROBLEMS

1. Volume, $R = 3$

ANSWER $36\pi$

2. Volume, $R = 6$

ANSWER $288\pi$

3. Volume, $R = 1$

ANSWER $\frac{4\pi}{3}$

4. Volume of a ball of diameter 10

ANSWER $\frac{500\pi}{3}$

5. Hemisphere, $R = 3$

ANSWER $18\pi$

6. A ball is what fraction of its cylinder?

ANSWER $\frac{2}{3}$

7. Volume, $R = 10$

ANSWER $\frac{4000\pi}{3}$

## Medium · the standard question

7 PROBLEMS

1. Ball of diameter 12: its volume

   ANSWER $288\pi$

2. Cap of height 4 in a sphere of radius 10

   ANSWER $\frac{416\pi}{3}$

3. Ball $R = 3$ in a cylinder $r = 5$: the rise

   ANSWER $1.44$ cm

4. A ball of volume $972\pi$: its radius

   ANSWER $9$

5. A hemisphere of radius 6: volume and curved area

   ANSWER $144\pi$, $72\pi$

6. A cap of height 2 in a sphere of radius 5

   ANSWER $\frac{52\pi}{3}$

7. Two balls of radii 3 and 6: the volume ratio

   ANSWER $1 : 8$

## Hard · stretch and reason

7 PROBLEMS

1. A sphere of radius 10 cut by planes at $4$ and $7$ from the centre: the layer volume

   ANSWER $\approx 314$

2. A ball just fits in a cube of edge 12: the unused volume

   ANSWER $1728 - 288\pi \approx 823$

3. Derive $V = \dfrac{4}{3}\pi R^3$ by integration

   ANSWER $\pi\int(R^2 - x^2)dx$ over $[-R, R]$

4. A spherical cap of volume $\dfrac{16\pi}{3}$ in a sphere of radius 4: its height

   ANSWER $2$

5. A ball of radius $r$ floats with half submerged. Its density relative to water

   ANSWER $0.5$

6. A hemisphere of radius 6 filled with water is poured into a cone $r = 6$. The depth

   ANSWER $12$

7. Three balls of radius 2 in a cylinder $r = 2, h = 12$: the empty fraction

   ANSWER $\dfrac{1}{3}$

## 07 Homework

### Homework — 5 tasks

Halve the diameter before cubing, every time.

1. Find the volume of a ball of radius $7$ cm and of one of diameter $9$ cm.
2. A cap of height $3$ is cut from a sphere of radius $12$. Find its volume.
3. A ball of radius $4$ is dropped into a cylinder of radius $6$. Find the rise in the level.

4. A ball has volume $\frac{2048\pi}{3}$. Find its radius and its surface area.

5. Explain in three sentences why a sphere is exactly two thirds of its enclosing cylinder.